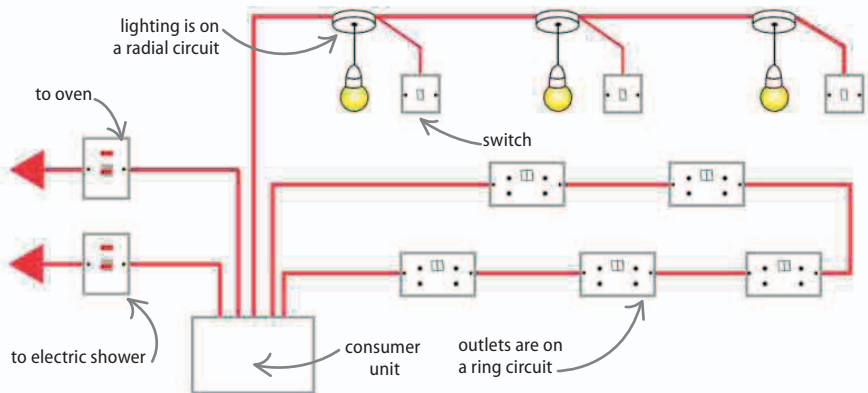


Domestic circuits

A domestic electricity supply connects to the grid at the consumer unit, or fuse box. Powerful electrical appliances, such as an oven, have a direct connection to the fuse box. Others are connected by ring or radial circuits. Ring circuits can use thinner wires to supply the same power as a radial circuit, but radial circuits can be extended easily and can carry smaller amounts of current. So, radial circuits are often used for lighting and ring circuits are used for power sockets.



△ Wiring the house

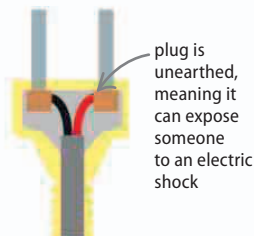
This simplified diagram shows the wiring in a house. Normally each floor of a house has two circuits: one for the lighting, another for the electrical outlets.

Protecting circuits

If too much current runs through domestic circuits, the wiring or appliances connected to it may get very hot and cause a fire. The consumer unit contains automatic switches called circuit breakers that cut off the supply if dangerous electrical surges occur. The circuit breaker also responds to short circuits, where faulty wiring or a damaged appliance results in the circuit drawing much more current than is normal. Fuses in plugs will also cut dangerous currents.

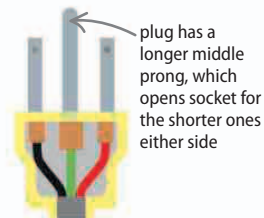
▽ Electrical plug

Most appliances connect to the electrical supply via a plug that fits into an outlet in the wall. Every plug has a live wire that delivers the current to it. The neutral wire carries the current back to the main circuit.



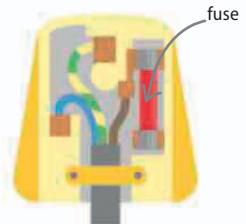
▽ Earth wire

A third wire is sometimes used in plugs. The earth wire connects the appliance to the ground via the domestic circuit. If a fault damages the insulation in the plug, any leaking current will flow safely to the ground via the earth wire.



▽ Fused plugs

The plugs in many countries are fitted with fuses—thin wires through which the current passes. If too much current passes through, the fuse wire gets hot and melts, breaking the circuit before other components get too hot.



Adaptor

Many devices come equipped with an oversized plug, known as an adaptor. The current flowing through domestic circuits is AC, which is fine for simple devices such as light bulbs and heaters. However, the back-and-forth surges of an AC supply would damage sensitive electronics, such as microchips, so an adaptor is used to filter the AC into a direct current (DC), which only travels in one direction.

▷ Rectifier

The main component in an adaptor is the rectifier. This is a type of diode (D) that only lets current pass in one direction.

